《操作系统综合实验》 实验报告

实验名称	Linux 内核编译	实验序号	7
实验日期	2025-05-29	实验人	魏朝旭

一、 实验题目

下载、编译内核源代码

启动测试所编译出来的内核

二、 相关原理与知识

1、linux 内核的相关知识

2、Vmare 虚拟机的使用

3、内核安装的一些依赖工具和相关命令

三、 实验过程

1、更新软件包列表并安装编译工具和依赖

sudo apt-get update

sudo apt-get install build-essential libncurses-dev bison flex libssl-dev libelf-dev wget

1）build-essential 是一个包含了编译软件所需的核心开发工具的软件包。它通常包括 gcc 编译器、用于 C++的 g++、make 以及其他必要的工具。

2）libncurses-dev 是 ncurses 库的开发包，用于创建类 Unix 系统中的基于文本的用户界面。

3）bison 是一个解析器生成器，它可以将语法描述转换为解析和翻译源代码或数据的 C 程序。

4）flex 是一个词法分析器生成器，它根据一组正则表达式生成 C 源代码。

5）libssl-dev 是 SSL（安全套接字层）和 TLS（传输层安全性）库的开发包，用于实现互联网上的安全通信。

6）libelf-dev 是 ELF（可执行和可链接格式）库的开发包，用于处理 ELF 文件，这种格式通常用于 Unix 和类 Unix 系统的可执行文件、目标代码、共享库和核心转储。

7）wget 是一个命令行工具，用于从网络下载文件。它支持 HTTP、HTTPS 和 FTP 协议。

```
┌──(liao㉿kali)-[~]
└─$ sudo apt-get update
[sudo] password for liao:
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [19.9 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [47.3
MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [115 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [27
1 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [192 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [8
62 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages
 [33.1 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents
 (deb) [16.9 kB]
Fetched 68.7 MB in 15s (4478 kB/s)
Reading package lists... Done

┌──(liao㉿kali)-[~]
└─$ sudo apt-get install build-essential libncurses-dev bison flex libssl-dev libelf-dev wget bc kmod cpio
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10).
build-essential set to manually installed.
flex is already the newest version (2.6.4-8.2+b2).
flex set to manually installed.
cpio is already the newest version (2.15+dfsg-1).
The following packages were automatically installed and are no longer required:
  libnsl-dev libtirpc-dev
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  gnutls-bin libasound2-data libblkid1 libc-bin libc-dev-bin libc-devtools libc-l10n libc6 libc6-dev libc6-i386 libdw1t64 libelf1t64
  libevent-core-2.1-7t64 libevent-openssl-2.1-7t64 libevent-pthreads-2.1-7t64 libgnutls-dane0t64 libgnutls30t64 libhogweed6t64 libkmod2
  libmount1 libncurses6 libncursesw6 libnettle8t64 libpsl5t64 libreadline8t64 libssl3t64 libtinfo6 libzstd-dev locales ncurses-base
  ncurses-bin ncurses-term openssl readline-common tmux
Suggested packages:
  bison-doc alsa-utils glibc-doc libnss-nis libnss-nisplus ncurses-doc libssl-doc readline-doc
The following packages will be REMOVED:
  libdw1 libelf1 libevent-core-2.1-7 libevent-openssl-2.1-7 libevent-pthreads-2.1-7 libgnutls-dane0 libgnutls30 libhogweed6 libnettle8
  libpsl5 libreadline8 libssl3
The following NEW packages will be installed:
  bc bison libdw1t64 libelf-dev libelf1t64 libevent-core-2.1-7t64 libevent-openssl-2.1-7t64 libevent-pthreads-2.1-7t64 libgnutls-dane0t64
  libgnutls30t64 libhogweed6t64 libnettle8t64 libpsl5t64 libreadline8t64 libssl-dev libssl3t64 libzstd-dev
The following packages will be upgraded:
  gnutls-bin kmod libasound2-data libblkid1 libc-bin libc-dev-bin libc-devtools libc-l10n libc6 libc6-dev libc6-i386 libkmod2 libmount1
  libncurses-dev libncurses6 libncursesw6 libtinfo6 locales ncurses-base ncurses-bin ncurses-term openssl readline-common tmux wget
25 upgraded, 17 newly installed, 12 to remove and 1624 not upgraded.
Need to get 28.8 MB of archives.
After this operation, 20.9 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 ncurses-bin amd64 6.5-2 [433 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libc-devtools amd64 2.38-13 [58.1 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libc-dev-bin amd64 2.38-13 [50.3 kB]
Get:5 http://kali.download/kali kali-rolling/main amd64 libc6-dev amd64 2.38-13 [1917 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 libc6-i386 amd64 2.38-13 [2479 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 locales all 2.38-13 [3909 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 libc6 amd64 2.38-13 [2775 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 ncurses-base all 6.5-2 [270 kB]
Get:11 http://kali.download/kali kali-rolling/main amd64 ncurses-term all 6.5-2 [513 kB]
Get:13 http://http.kali.org/kali kali-rolling/main amd64 libelf1t64 amd64 0.191-1+b1 [189 kB]
Get:14 http://kali.download/kali kali-rolling/main amd64 libevent-pthreads-2.1-7t64 amd64 2.1.12-stable-10 [53.9 kB]
Get:15 http://kali.download/kali kali-rolling/main amd64 libevent-openssl-2.1-7t64 amd64 2.1.12-stable-10 [60.8 kB]
Get:16 http://kali.download/kali kali-rolling/main amd64 libevent-core-2.1-7t64 amd64 2.1.12-stable-10 [131 kB]
Get:17 http://kali.download/kali kali-rolling/main amd64 openssl amd64 3.2.1-3 [1360 kB]
Get:18 http://http.kali.org/kali kali-rolling/main amd64 kmod amd64 32+20240327-1 [96.8 kB]
Get:19 http://http.kali.org/kali kali-rolling/main amd64 libkmod2 amd64 32+20240327-1 [60.9 kB]
Get:20 http://kali.download/kali kali-rolling/main amd64 libssl3t64 amd64 3.2.1-3 [2244 kB]
Get:22 http://kali.download/kali kali-rolling/main amd64 wget amd64 1.24.5-1 [979 kB]
Get:23 http://kali.download/kali kali-rolling/main amd64 libnettle8t64 amd64 3.9.1-2.2 [296 kB]
Get:24 http://kali.download/kali kali-rolling/main amd64 libgnutls-dane0t64 amd64 3.8.5-4 [435 kB]
Get:25 http://kali.download/kali kali-rolling/main amd64 libhogweed6t64 amd64 3.9.1-2.2 [328 kB]
```

2、下载内核源代码并解压缩

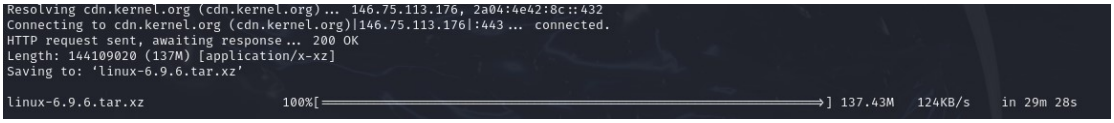
cd /usr/src

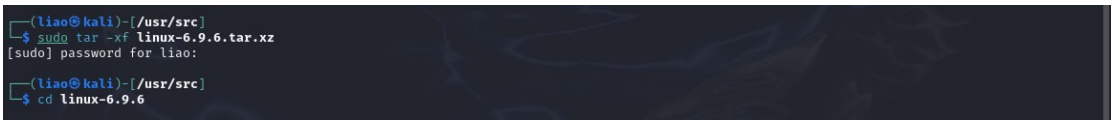
sudo wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.9.6.tar.xz

sudo tar -xf linux-6.9.6.tar.xz

cd linux-6.9.6







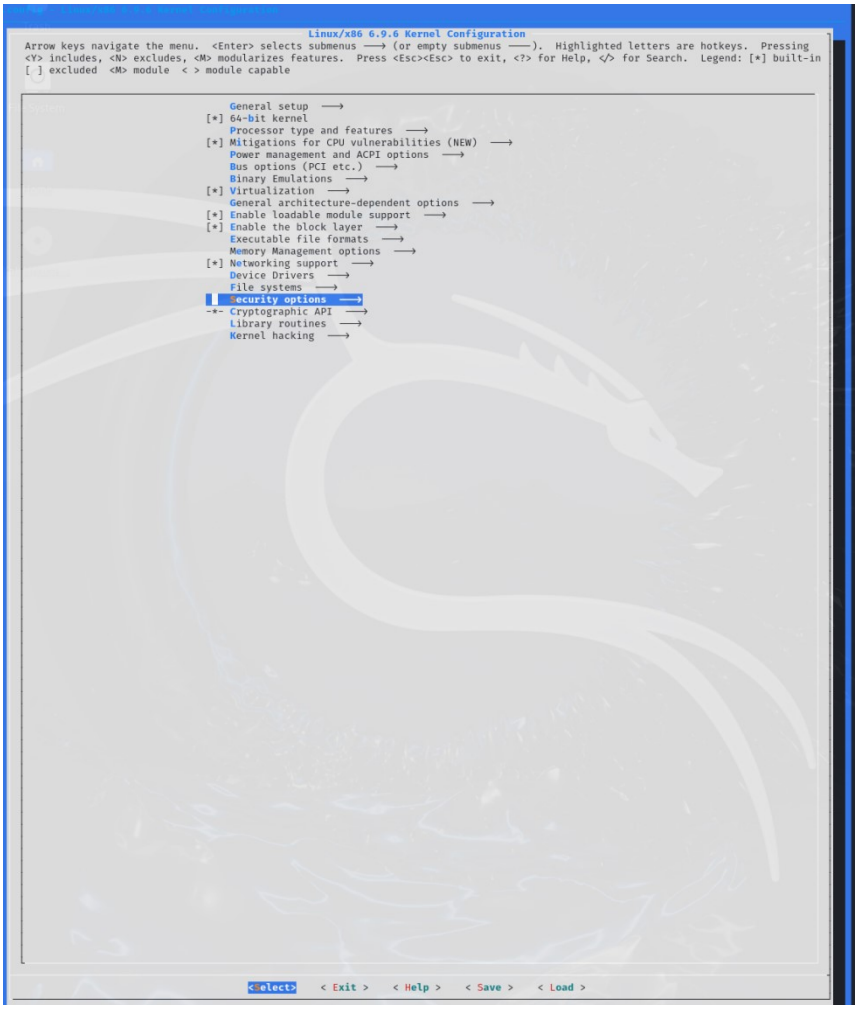
3、复制当前内核配置并配置内核选项，保存退出即可

sudo cp /boot/config-$(uname -r) .config

sudo chown $USER:$USER .config

make menuconfig





4、编译内核和内核模块

Sudo make -j$(nproc)

等待时间长一点

部分截图



sudo make modules_install

5、安装编译好的内核

Sudo make install

6、更新 Grub 引导配置并重启

sudo update-grub

sudo reboot

7、验证是否成功启动内核

uname -r

输出显示新内核版本号

四、 实验结果与分析

在本次实验中，成功地下载了 Linux 内核源码并完成编译的工作，同时还完成了编译内核模块与安装新内核的工作

重新进入系统，我们可以看到我们的内核版本已经被替换为 6.9.6。

```
liao@kali:~/usr/src/linux-6.9.6$ uname -a
Linux kali 6.9.6 #1 SMP Wed Jun 23 15:45:46 PDT 2024 x86_64 GNU/Linux
```

五、 问题总结

1、在进行到第 5 步编译内核模块的时候会出现问题，查找资料只能重新去编译内核，删除之前的编译内容，最后发现是之前的依赖软件没有安装完全。

Sudo make clean

Sudo make mrproper

重新安装一些依赖就可以了

六、 源代码

```
sudo apt-get update
sudo apt-get install build-essential libncurses-dev bison flex libssl-dev libelf-dev wget
cd /usr/src
sudo wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.9.6.tar.xz
sudo tar -xf linux-6.9.6.tar.xz
cd linux-6.9.6
sudo cp /boot/config-$(uname -r) .config
sudo chown $USER:$USER .config
make menuconfig
```

```
Sudo make -j$(nproc)
sudo make modules_install
Sudo make install
 sudo update-grub
sudo reboot
uname -r
```